FINAL PROJECT REPORT

# E-Commerce Delivery And Shipping Performance Analysis Using Python

*From order data to operational insight*

Python • Pandas • NumPy • Exploratory Data Analysis • Visualization

**Prepared for:** PROGRAME IN AI DRIVEN DATA ANALYTICS FINAL PROJECT

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# Executive Summary

This project analyzes 50,000 e-commerce orders to understand delivery performance, shipping cost, delay patterns, customer experience, return behavior, and carrier performance. The analysis follows a complete Python workflow covering data loading, quality checks, preprocessing, feature engineering, exploratory data analysis (EDA), statistical summaries, groupby and pivot analysis, correlation analysis, and visualization.

The dataset contained 30 original columns. The only substantial missing field was return_reason, with 41,375 blanks; these were validated as contextual because they occurred when no return was requested and were replaced with "Not Applicable". No duplicate records were found. Date-based features for order month and order day were then engineered, resulting in a final analysis-ready dataset of 50,000 rows × 32 columns.

The completed analysis found that International shipping had the highest average delay and cost, while Same Day had the lowest average delay and the highest average customer rating. Distance was strongly positively associated with shipping cost (r = 0.82), while delivery delay was moderately negatively associated with customer rating (r = -0.51). Return requests accounted for approximately 17.2% of orders, with Product Defect as the most common return reason.

| 50,000 | 32 | 12 | 11 |
|---|---|---|---|
| ORDERS | FINAL FEATURES | VISUALIZATIONS | KEY FINDINGS |

## Report Structure

- Project background and objectives
- Dataset overview and methodology
- Data cleaning and preprocessing
- Exploratory data analysis and visual findings
- Key business insights and recommendations
- Conclusion and future improvement opportunities

# 1. Project Background and Objectives

## 1.1 Business Context

E-commerce delivery performance directly influences customer experience and operational efficiency. Orders move through warehouses, carriers, shipping methods, locations, and changing delivery conditions. Delays, shipping cost, returns, and customer satisfaction are interconnected, making delivery data suitable for data-driven operational analysis.

## 1.2 Problem Statement

**How can e-commerce order and delivery data be used to understand shipping performance, delays, costs, returns, and customer satisfaction?**

## 1.3 Project Objectives

- Measure overall delivery performance.
- Identify factors associated with delivery delays.
- Compare shipping methods and carrier performance.
- Examine shipping cost and distance patterns.
- Study customer ratings and return behavior.
- Generate practical business insights and recommendations.

## 1.4 Why Python Was Used

Python provided an end-to-end workflow for data loading, cleaning, transformation, exploratory analysis, statistical summaries, and visualization. The notebook used Pandas and NumPy for data handling and Matplotlib/Seaborn for visual analysis.

# 2. Dataset Overview

| 50,000 | 30 | 29 | 1 |
|---|---|---|---|
| RECORDS | ORIGINAL COLUMNS | COMPLETE COLUMNS | CONTEXTUAL MISSING FIELD |

The dataset combines customer, order, logistics, delivery, cost, experience, and return information in one table. This allows the project to examine operational performance from multiple business perspectives. The source dataset was obtained from Kaggle and used in the project as E-commerce_Delivery_Shipping_Data_2026.csv.

| Data Area | Representative Variables |
|---|---|
| **Order** | order_id, order_date, order_value_usd, order_priority |
| **Customer** | customer_id, customer_segment, customer_city, customer_country |
| **Product** | product_category, product_weight_kg, package_size |
| **Shipping** | shipping_method, carrier, distance_km, shipping_cost_usd |
| **Delivery** | promised_delivery_days, actual_delivery_days, delivery_delay_days, delivery_status |
| **Operating Conditions** | weather_condition, warehouse_processing_hours, delivery_attempts |
| **Customer Experience** | customer_rating, return_requested, return_reason |

## 2.1 Initial Inspection

The notebook first reviewed dataset shape, column names, data types, descriptive statistics, missing values, and duplicate records. This established the quality baseline before any analysis was performed.

# 3. Methodology and Data Preparation

## 3.1 Analysis Workflow

1. Load and understand the dataset
2. Inspect data quality
3. Handle contextual missing values
4. Check duplicate records
5. Correct date data type
6. Engineer month/day features
7. Validate final data
8. Perform univariate, bivariate, and multivariate EDA
9. Create visualizations and statistical comparisons
10. Summarize findings and recommendations

## 3.2 Missing Values and Duplicates

The return_reason field contained 41,375 missing values. A cross-tabulation between return_requested and missing return_reason values confirmed that all blanks occurred where no return was requested. The blanks were therefore treated as contextual and replaced with “Not Applicable”. No duplicate rows were identified, so no records required removal.

## 3.3 Data Type Correction and Feature Engineering

The order_date column was converted from text/object format to datetime. Two new features were then created: order_month and order_day. These supported monthly and weekday-based analysis. The final dataset contained 50,000 rows, 32 columns, zero missing values, and zero duplicates.

# 4. Exploratory Data Analysis

EDA was structured to move from simple distributions to operational comparisons and numerical relationships. The notebook included univariate frequency checks, groupby summaries, a pivot table, correlation analysis, and multiple visualizations.

## 4.1 Shipping Method Summary

| Shipping Method | Avg Delay (Days) | Avg Cost (USD) | Avg Rating |
|---|---|---|---|
| **International** | 2.08 | $204.40 | 3.20 |
| **Economy** | 1.50 | $48.91 | 3.28 |
| **Standard** | 0.96 | $71.55 | 3.40 |
| **Express** | 0.62 | $107.58 | 3.57 |
| **Same Day** | 0.29 | $136.58 | 3.84 |

International shipping stands out as the highest-delay and highest-cost method, while Same Day has the lowest average delay and the highest average customer rating. Economy is the least expensive method but has a higher average delay than Standard and Express.

## 4.2 Correlation Findings

- Promised delivery days ↔ actual delivery days: r = 0.93
- Distance ↔ shipping cost: r = 0.82
- Actual delivery days ↔ delivery delay: r = 0.71
- Delivery delay ↔ customer rating: r = -0.51

These coefficients describe association, not causation. They are used here as evidence of relationships that deserve operational attention.

# 5. Visual Findings: Volume and Shipping Performance

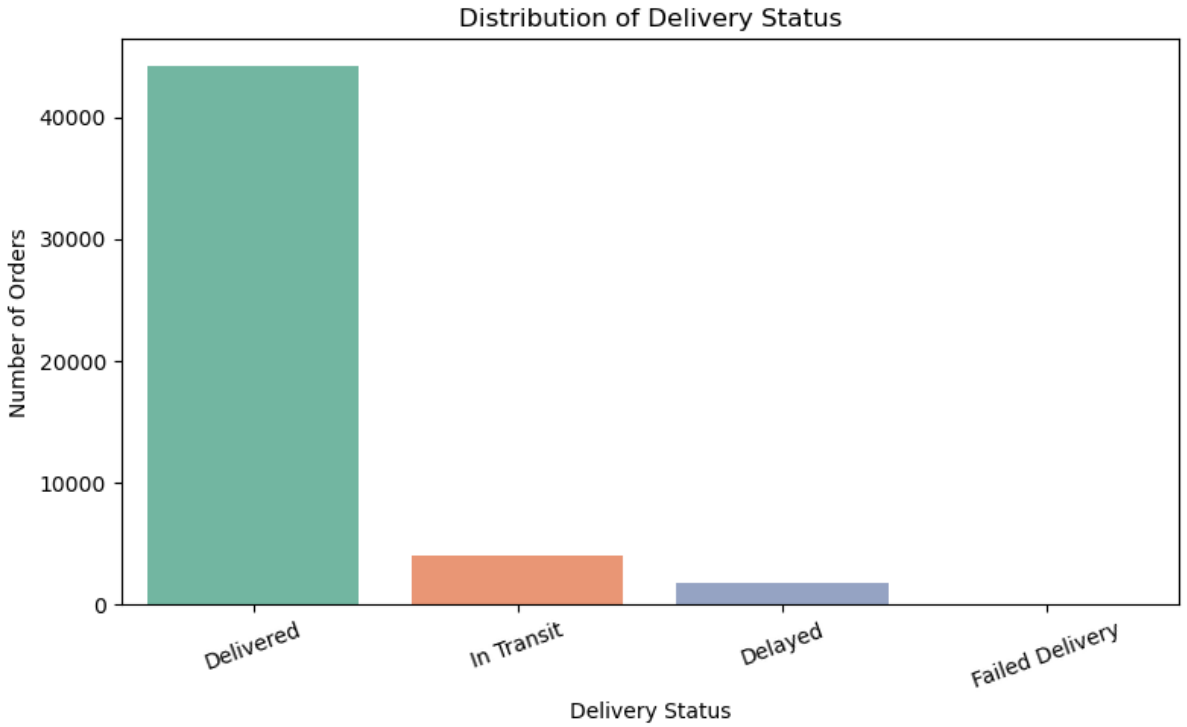



*Figure 1. Delivery status distribution*

Delivered orders dominate the dataset with 44,184 orders. In Transit accounts for 4,007 orders and Delayed for 1,806, while Failed Delivery is extremely rare at only 3 orders.




*Figure 2. Shipping method distribution*

Standard is the most frequently used shipping method (17,027 orders), followed by International (15,839). Same Day is the least frequently used method (1,861 orders).

# 6. Visual Findings: Delay, Cost and Weather




*Figure 3. Average delivery delay by shipping method*

International has the highest average delay at 2.08 days. Same Day has the lowest average delay at 0.29 days. Express and Standard perform between these two extremes, while Economy averages 1.50 days of delay.

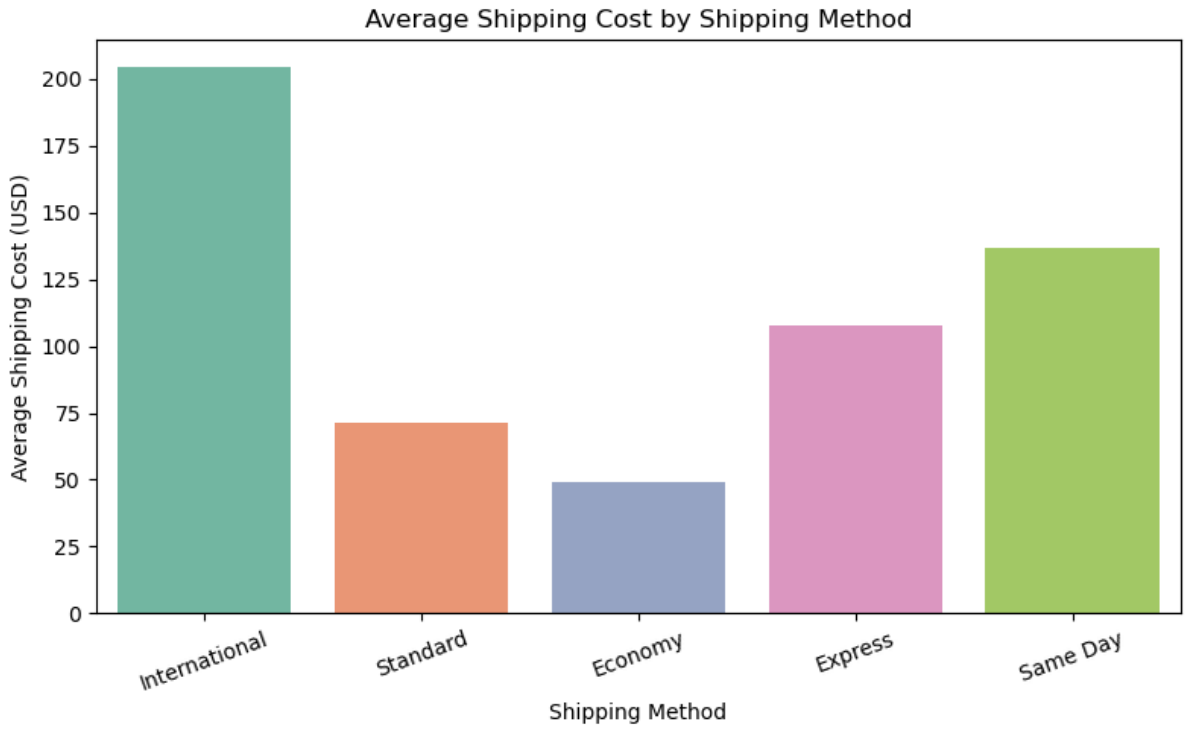



*Figure 4. Average shipping cost by shipping method*

International also has the highest average shipping cost at $204.40. Economy is the lowest-cost option at $48.91. This highlights the different speed-cost profiles across shipping methods.

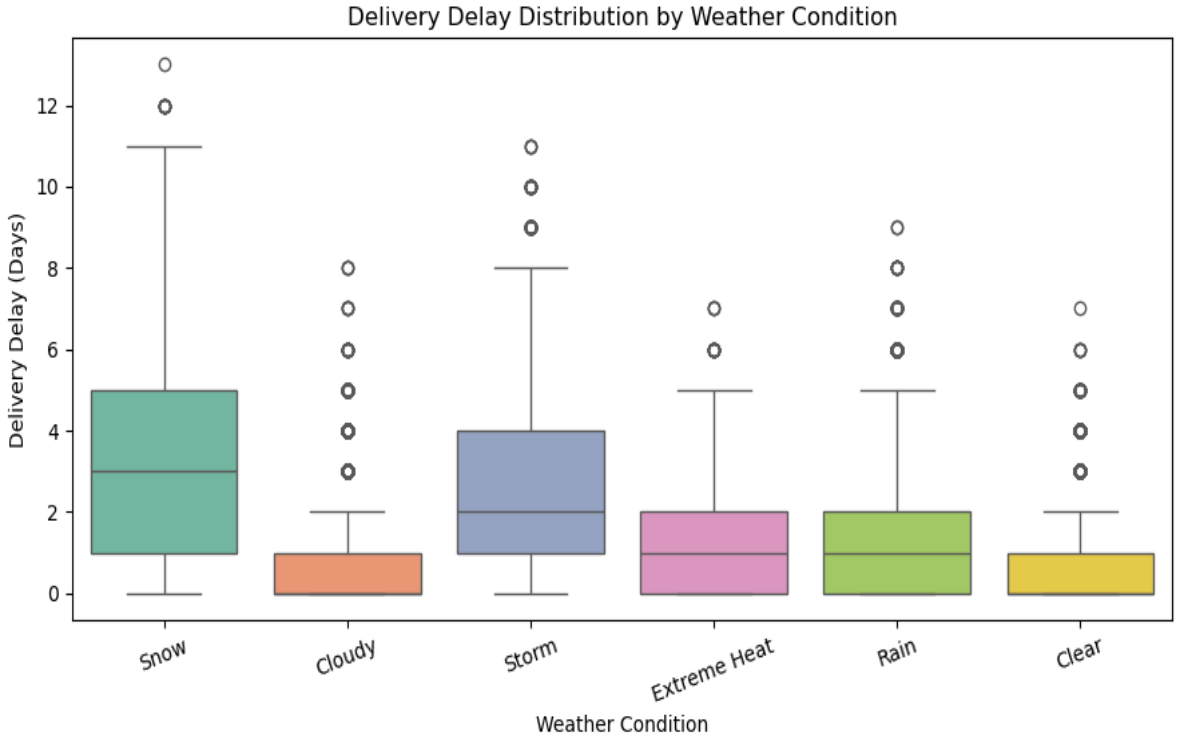



*Figure 5. Delivery delay distribution by weather condition*

Delivery delays vary noticeably across weather conditions. Snow shows the highest median delay and the greatest spread, followed by Storm, while Clear and Cloudy conditions generally show lower delays. Several high-delay outliers are also observed across the weather categories




*Figure 6. On-Time Delivery Rate by Shipping Method*

This visualization compares the on-time delivery performance of different shipping methods. A higher percentage indicates better delivery reliability. The analysis helps identify which shipping methods are more effective in meeting promised delivery times.

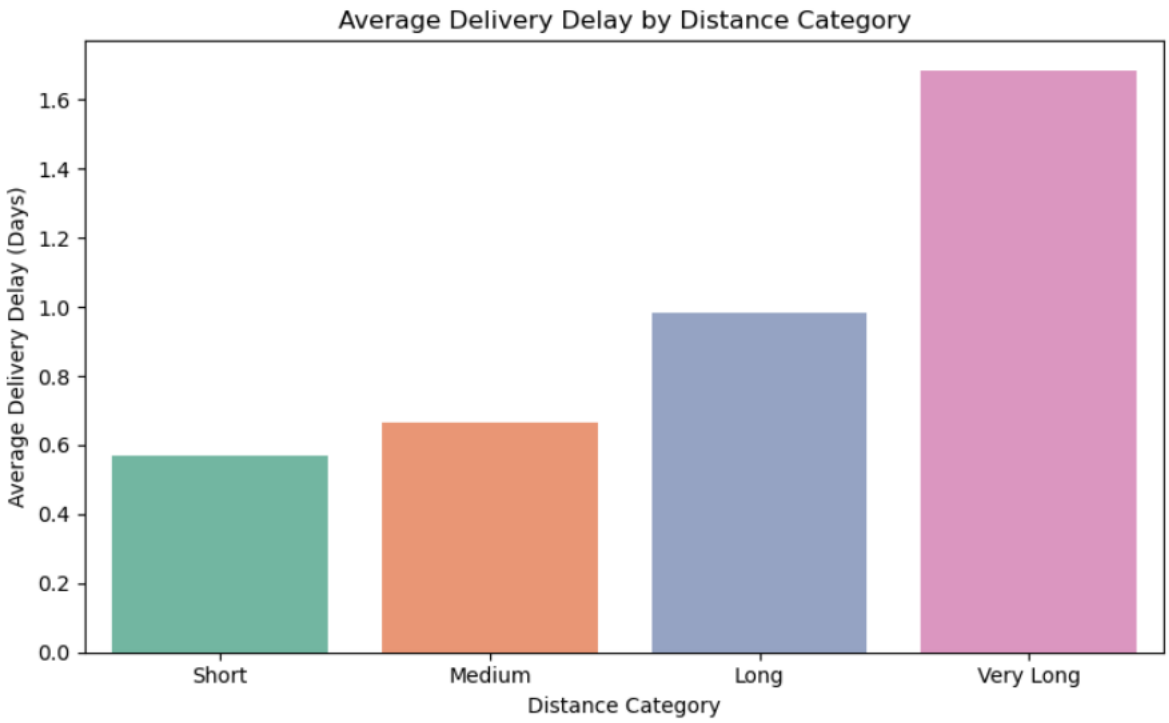



Figure 7. Average Delivery Delay by Distance Category

This visualization compares the average delivery delay across different distance categories. It helps determine whether delivery distance has an impact on delivery delays and identifies which distance category experiences higher delays

# 7. Visual Findings: Cost Drivers, Customer Experience and Trends

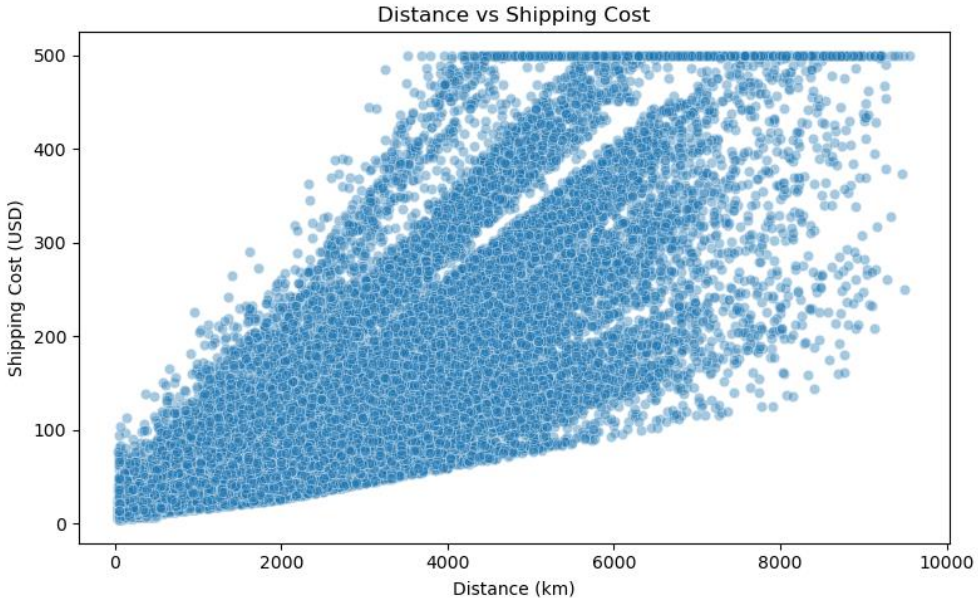



Figure 8. Distance vs shipping cost

Longer shipping distances are generally associated with higher shipping costs. The relationship is strong and positive (r = 0.82), making distance an important logistics cost driver in this dataset.

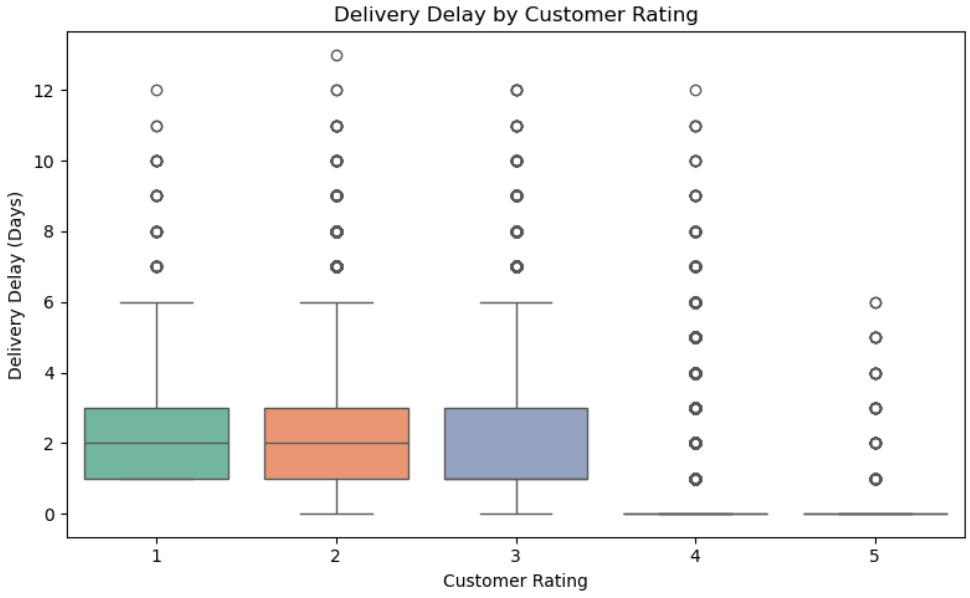



Figure 9. Delivery delay vs customer rating

Lower customer ratings are associated with higher delivery delays. Ratings 1–3 show higher typical delays, while ratings 4 and 5 generally have little or no delay. This pattern is consistent with the negative correlation (-0.51) between delivery delay and customer rating.

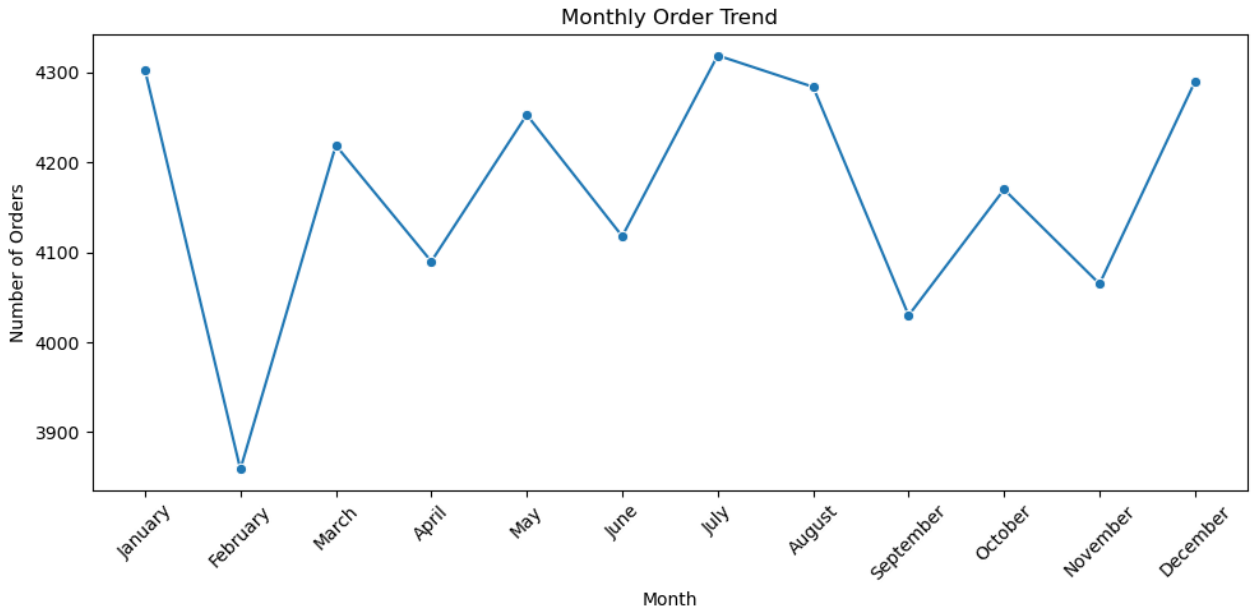



*Figure 10. Monthly order trend*

Monthly order volume remains relatively stable throughout the year with moderate fluctuations. July records the highest number of orders, while February records the lowest. No strong overall increasing or decreasing trend is observed.

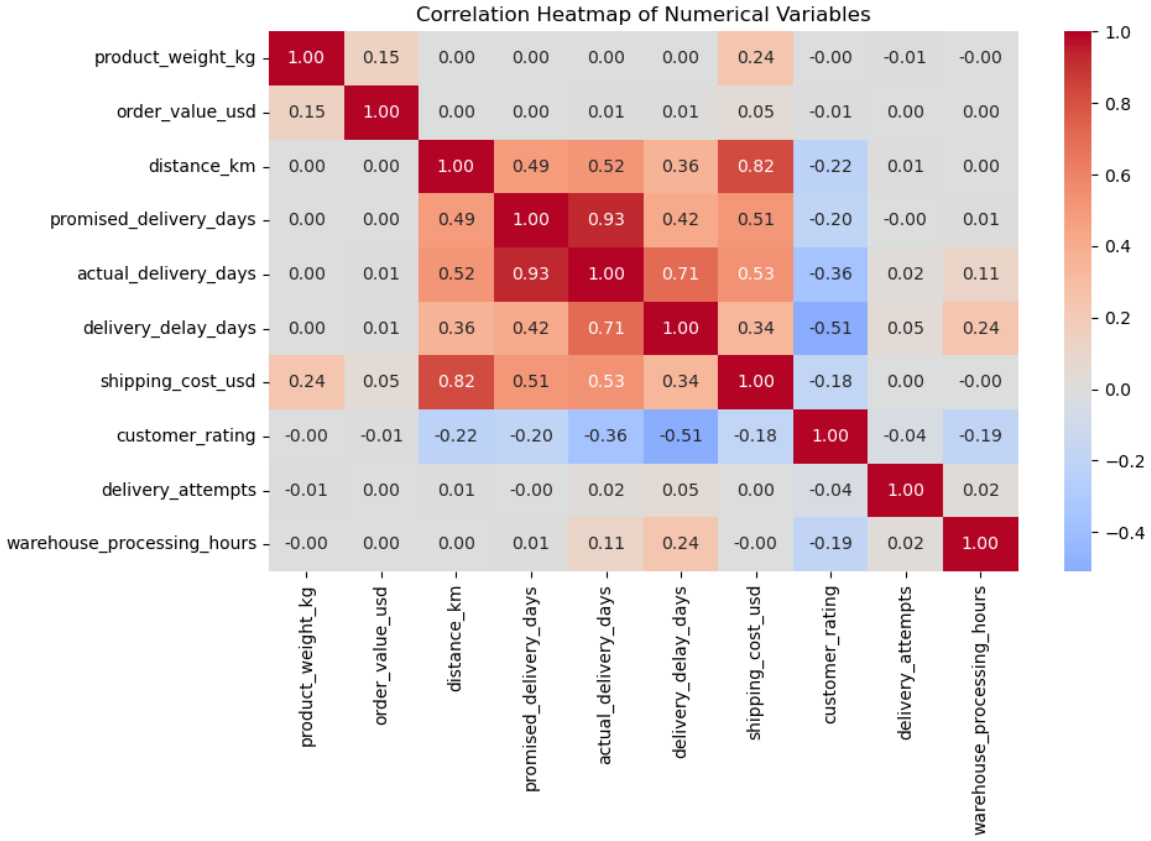



*Figure 11. Correlation heatmap*

The heatmap shows strong positive relationships between promised and actual delivery days (0.93) and between distance and shipping cost (0.82). Actual delivery days and delivery delay are also strongly positively related (0.71). Delivery delay has a moderate negative relationship with customer rating (-0.51).

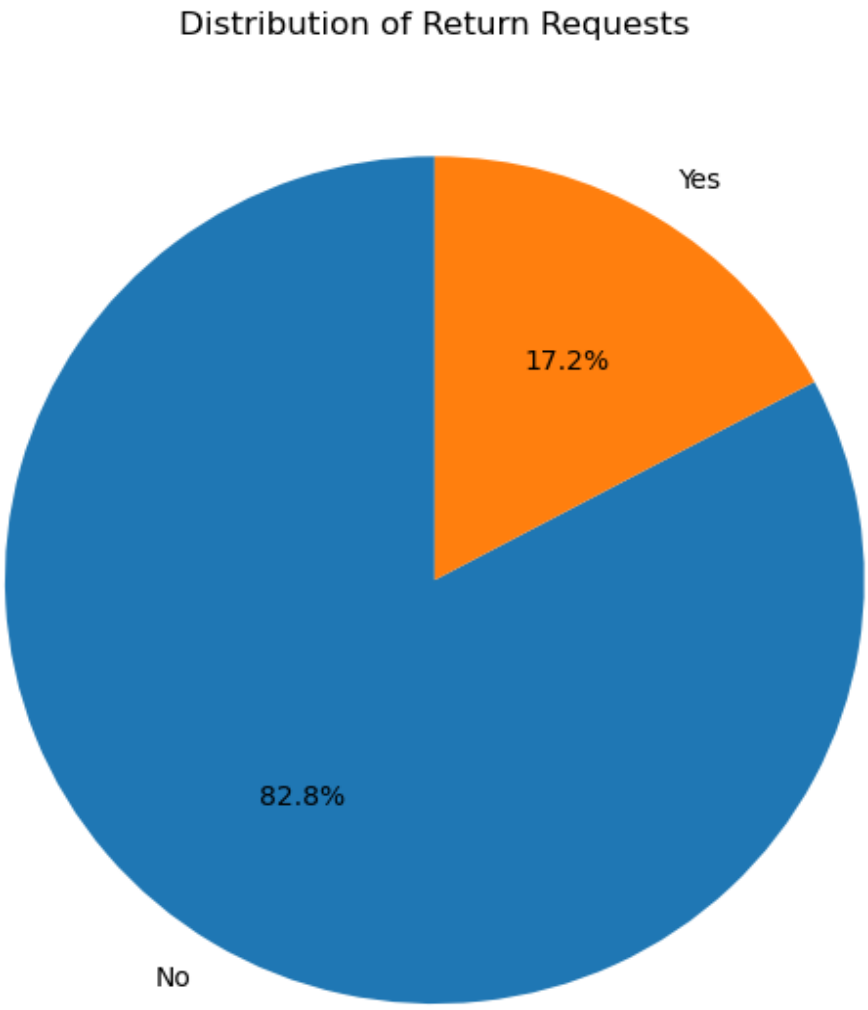



*Figure 12. Return request distribution*

The majority of orders (82.8%) did not have a return request, while 17.2% of orders had a return request. Return requests therefore represent a relatively smaller proportion of total orders.

# 8. Visual Findings: Returns and Carrier Performance

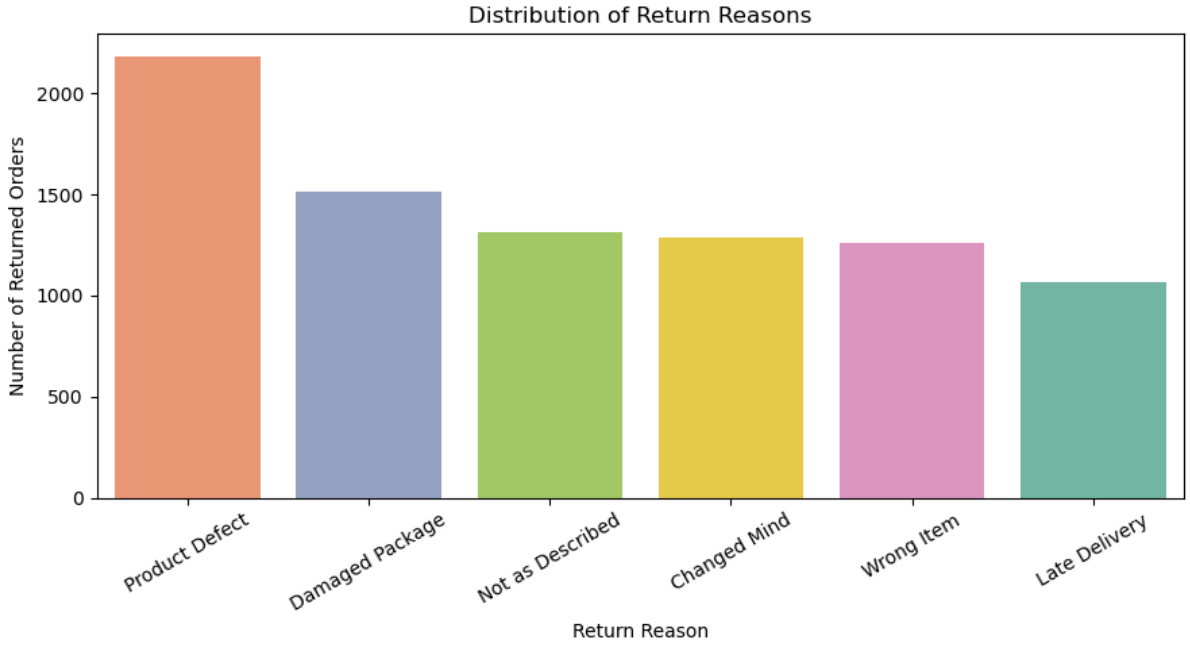



Figure 13. Return reasons distribution

Return requests represent approximately 17.2% of orders. Product Defect is the leading return reason, followed by Damaged Package. Late Delivery is the least common return reason among returned orders.

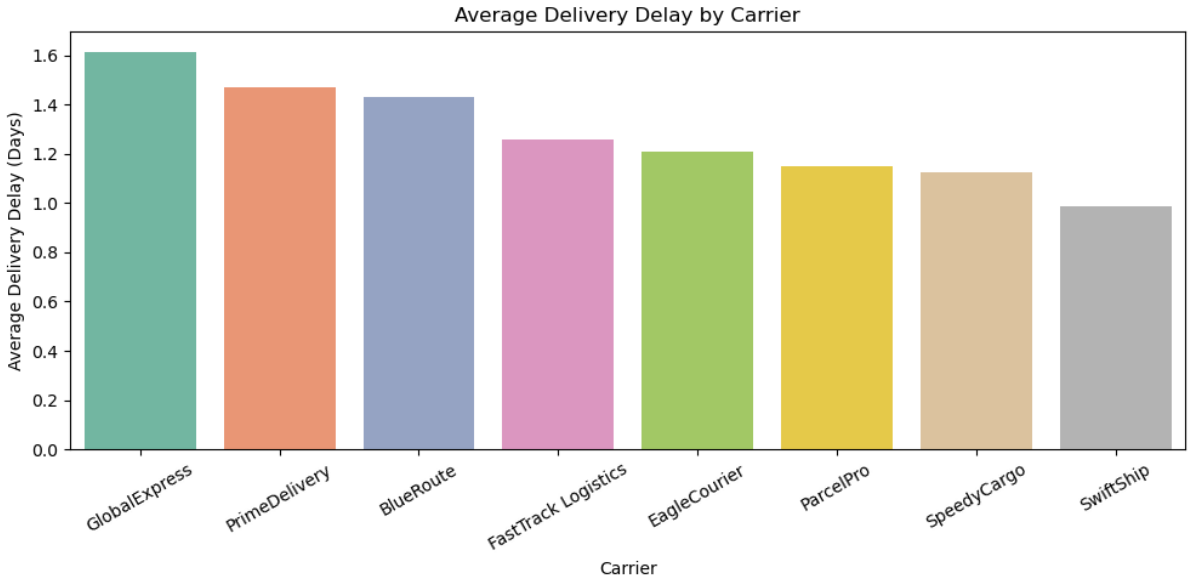



Figure 14. Average delivery delay by carrier

Average delivery delay varies across carriers. GlobalExpress shows the highest average delay at approximately 1.61 days, followed by PrimeDelivery and BlueRoute. SwiftShip records the lowest average delay at approximately 0.99 days.

# 9. Additional Operational Findings

The Jupyter Notebook also provides operational views on weather conditions and monthly order patterns, complementing the shipping-method, cost, customer-experience, return, and carrier analyses.

## 9.1 Shipping Method Performance

Same Day records the lowest average delivery delay (0.29 days) and the highest average customer rating (3.84), while International records the highest average delay (2.08 days) and highest average shipping cost ($204.40).

## 9.2 Distance and Shipping Cost

Delivery distance has a strong positive relationship with shipping cost (r = 0.82), showing that longer delivery distances are generally associated with higher shipping costs.

## 9.3 Weather Conditions

Snow shows the highest typical delivery delays and widest spread, followed by Storm. Clear and Cloudy conditions generally show lower delays. This suggests that adverse weather should be considered in delivery planning and contingency measures.

## 9.4 Monthly Order Pattern

Monthly order volume remains relatively stable across the year. July records the highest number of orders and February the lowest, but no strong overall upward or downward trend is observed.

# 10. Key Findings

1. Standard is the most frequently used shipping method; Same Day is the least used.
2. International has the highest average delay (2.08 days) and average shipping cost ($204.40).
3. Same Day has the lowest average delay (0.29 days) and the highest average customer rating (3.84).
4. Snow and Storm conditions are associated with higher delivery delays compared with other weather conditions.
5. Delivery distance has a strong positive relationship with shipping cost (r = 0.82).
6. Delivery delay has a moderate negative relationship with customer rating (r = -0.51).
7. Monthly order volumes remain relatively stable, with July recording the highest volume and February the lowest.
8. Approximately 17.2% of orders have a return request; Product Defect is the most common return reason.
9. Carrier performance varies, with GlobalExpress showing the highest average delivery delay and SwiftShip the lowest.
10. Promised and actual delivery days have a strong positive relationship (r = 0.93).
11. Actual delivery days and delivery delay are strongly positively related (r = 0.71).

## Interpretation

Taken together, the findings show that delivery performance is not driven by a single factor. Shipping method, distance, weather, and carrier performance all show meaningful patterns, while delay is also associated with customer experience. The return analysis adds a separate quality dimension, showing that product defects and package damage are more prominent return drivers than late delivery.

# 11. Business Recommendations

- **International shipping:** Optimize international shipping operations to reduce delay and improve on-time performance while reviewing the drivers of higher shipping cost.
- **Time-sensitive orders:** Prioritize Same Day or Express options where faster delivery is operationally appropriate.
- **Weather planning:** Strengthen contingency planning for Snow and Storm conditions, which are associated with higher delays.
- **Distance-aware logistics:** Consider delivery distance in shipping-cost and logistics planning because longer distances are strongly associated with higher shipping costs.
- **Customer satisfaction:** Reduce avoidable delivery delays to support stronger customer ratings and service experience.
- **Return reduction:** Improve product quality checks and packaging controls because Product Defect and Damaged Package are leading return reasons.
- **Carrier management:** Review carrier delay performance regularly and investigate the operational causes behind higher-delay carriers.

# 12. Conclusion

This project demonstrates a complete Python-based data analytics workflow using a large e-commerce delivery dataset. Starting from 50,000 raw order records, the analysis validated data quality, handled contextual missingness, corrected date formatting, engineered useful time features, and performed structured exploratory analysis.

The analysis identified clear patterns in shipping speed, shipping cost, delivery distance, weather exposure, customer ratings, returns, and carrier performance. International shipping showed the strongest delay and cost challenges, while Same Day performed strongly on speed and customer rating. Distance emerged as a major cost driver, and delivery delay was associated with lower customer ratings. Product defects and damaged packages were important return drivers.

Overall, the project shows how data-driven analysis can support delivery efficiency, cost management, customer satisfaction, return reduction, and carrier review. The findings provide a practical foundation for further dashboarding, operational monitoring, and predictive analysis.

| 50,000 | 32 | 12 | 11 |
|---|---|---|---|
| CLEAN RECORDS | FINAL FEATURES | VISUALIZATIONS | EVIDENCE-BASED FINDINGS |

## Future Scope

- Build an interactive Power BI dashboard for operational monitoring.
- Add predictive modeling for late-delivery risk.
- Analyze warehouse-level performance and processing bottlenecks.
- Develop carrier scorecards that combine delay, cost, and reliability metrics.
- Track changes over time using refreshed operational data.

# Appendix A. Tools and Project Deliverables

| Item | Use in Project |
|---|---|
| **Python** | Core analysis environment |
| **Pandas** | Data loading, cleaning, grouping, transformation |
| **NumPy** | Numerical support |
| **Matplotlib / Seaborn** | Data visualization |
| **Jupyter Notebook** | Structured analysis workflow and documentation |
| **PowerPoint** | Final project presentation and communication of findings |
| **CSV Dataset** | Kaggle source dataset containing 50,000 e-commerce order records |

# Appendix B. Core Analysis Techniques

- Descriptive statistics
- Missing-value validation
- Duplicate checking
- Date conversion and feature engineering
- Value counts and univariate analysis
- GroupBy aggregation
- Pivot-table analysis
- Correlation analysis
- Bar charts, line charts, scatter plots, box plots, pie chart, heatmap
- Evidence-based interpretation and business recommendations

## Project Source Files

This report was prepared from the Kaggle source dataset (E-commerce_Delivery_Shipping_Data_2026.csv) and the completed Jupyter Notebook provided for the project. All graphs shown in this report are taken directly from the final Jupyter Notebook outputs.